

Trainings ▾

Preparatory Steps

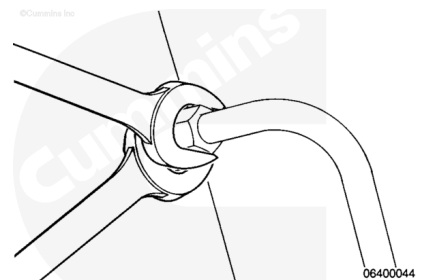
- Remove as much of the engine wiring harness and brackets as necessary to gain access to the fuel tubes. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/89/89-019-043.html>) in the Fuel System Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, and QSK60 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>).



Remove

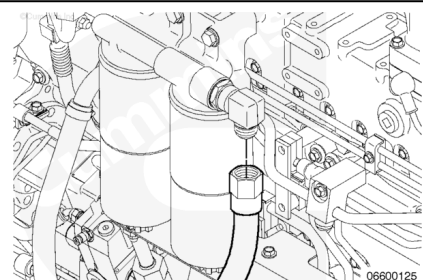
Use two wrenches to remove the fuel tubes and hoses.

Support the mating fittings with a wrench. Loosen the fuel tube nuts with the other wrench.



Tank to Fuel Filter

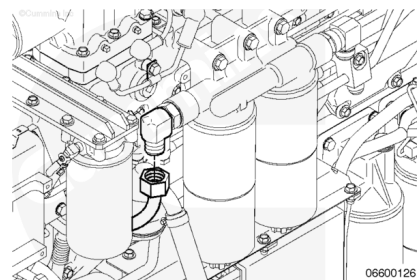
Remove the fuel filter inlet hose and related clamps.



Fuel Pump Inlet Line

Remove the fuel pump inlet line and line clamp.

Discard the o-rings.

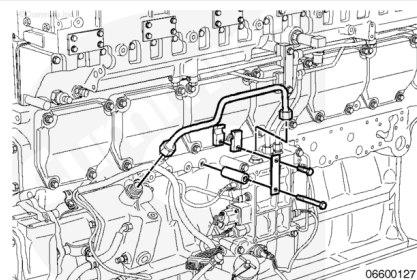


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Control Supply Line

Remove the fuel control supply line and line clamp.

Discard the o-rings.

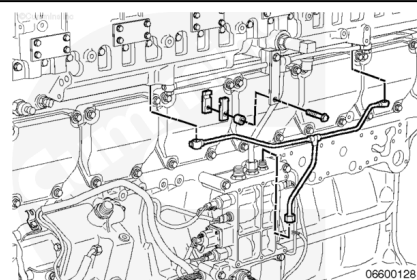


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Timing Supply Line

Remove the fuel timing supply line and line clamp.

Discard the o-rings and banjo fitting washers.

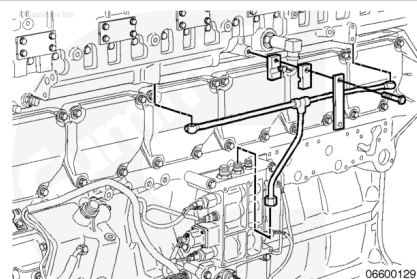


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Rail Supply Line

Remove the fuel rail supply line and line clamp.

Discard the o-rings and banjo fitting washers.



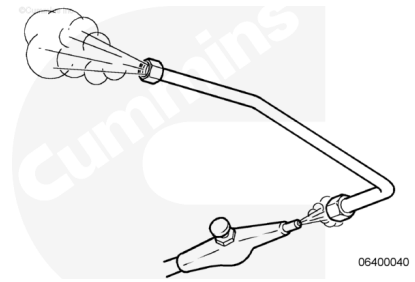
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Clean and Inspect for Reuse

⚠ WARNING ⚠

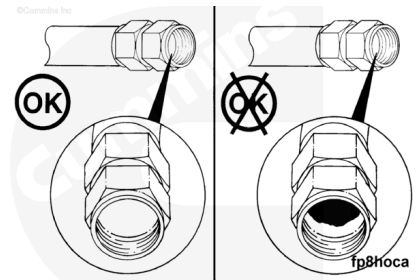
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use compressed air to flush the lines and remove any loose dirt particles.



Inspect the inside of the hose.

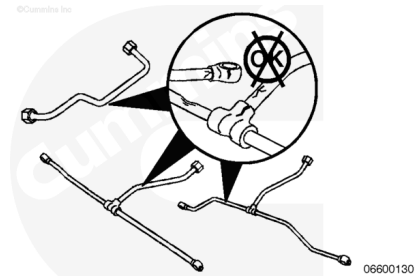
- The inner lining of the hose can separate from the center hose section.
- A separation of the inner lining can cause a restriction in the fuel flow.



Inspect for bends in the line that can restrict fuel flow.

Inspect the line for cracks that can cause a pressure loss.

Inspect the metal fuel lines for sharp bends that can cause a pressure restriction.

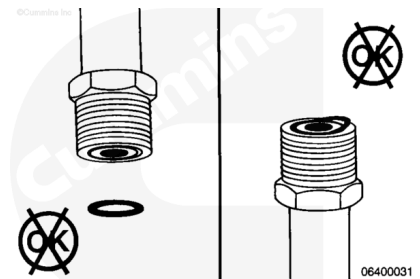


Install

Install the new o-rings in the fittings.

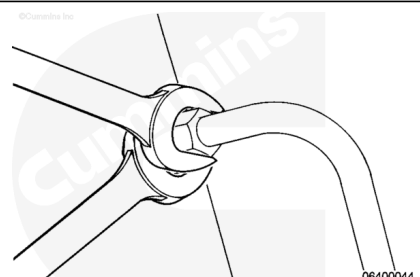
The fuel filter outlet, fuel pump inlet, fuel pump outlet, and fuel control module fittings use face seal o-rings.

Make sure the o-rings are properly seated during installation.



Use two wrenches to install the fuel tubes and hoses.

Support the mating fittings with a wrench. Tighten the fuel tube nuts with the other wrench.

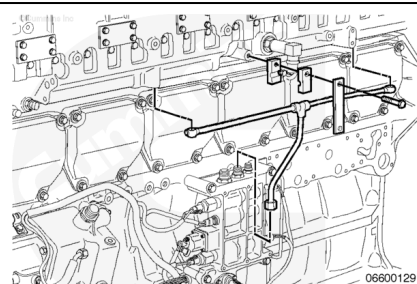


Rail Supply Line

Install a new o-ring in the rail supply line fitting and new washers on the banjo fittings.

Install the rail supply line.

Sleeve nut	27 mm	[20 ft-lb]
Banjo fitting	34 mm	[25 ft-lb]

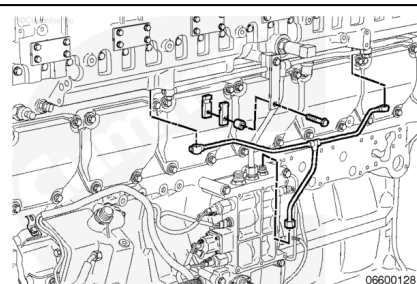


Timing Supply Line

Install a new o-ring in the timing supply line fitting and new washers on the banjo fittings.

Install the timing supply line and line clamp.

Sleeve nut	47 mm	[35 ft-lb]
Banjo fitting	44 mm	[32 ft-lb]

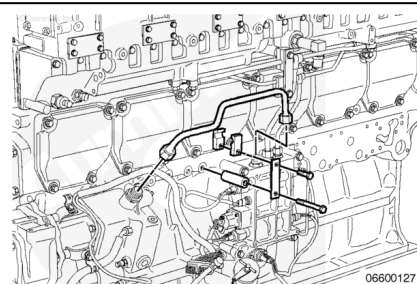


Control Supply Line

Install new o-rings in the fuel pump outlet and fuel control module inlet fittings.

Install the fuel control module supply line.

Torque Value: 68 n·m [50 ft-lb]

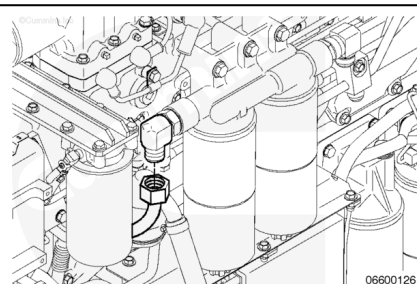


Fuel Pump Inlet Line

Install new o-rings in the fuel pump inlet and fuel control module outlet fittings.

Install the fuel pump inlet line and line clamp.

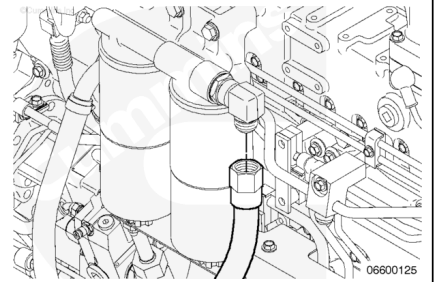
Torque Value: 95 n·m [70 ft-lb]



Tank to Fuel Filter

Install the fuel filter inlet line.

Torque Value: 95 n•m [70 ft-lb]



Finishing Steps

- Install the engine wiring harness and brackets, as necessary. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/89/89-019-043.html>) in the Fuel System Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, and QSK60 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>).

